

# Homework #2

Math 104: Intro to Analysis

Summer 2026

UC Berkeley

Donald Aingworth IV

July 10, 2026

## Contents

|          |                        |           |
|----------|------------------------|-----------|
| <b>1</b> | <b>Problem 1</b>       | <b>2</b>  |
| 1.1      | Solution (a) . . . . . | 2         |
| 1.2      | Solution (b) . . . . . | 3         |
| 1.3      | Solution (c) . . . . . | 4         |
| <b>2</b> | <b>Problem 2</b>       | <b>5</b>  |
| 2.1      | Solution (a) . . . . . | 6         |
| 2.2      | Solution (b) . . . . . | 7         |
| 2.3      | Solution (c) . . . . . | 8         |
| 2.4      | Solution (d) . . . . . | 9         |
| <b>3</b> | <b>Problem 3</b>       | <b>10</b> |
| 3.1      | Solution (a) . . . . . | 11        |
| 3.2      | Solution (b) . . . . . | 12        |
| 3.3      | Solution (c) . . . . . | 13        |
| <b>4</b> | <b>Problem 4</b>       | <b>14</b> |
| 4.1      | Solution (a) . . . . . | 15        |
| 4.2      | Solution (b) . . . . . | 16        |
| 4.3      | Solution (c) . . . . . | 17        |
| 4.4      | Solution (d) . . . . . | 18        |
| 4.5      | Solution (e) . . . . . | 19        |
| 4.6      | Solution (f) . . . . . | 20        |

# 1 Problem 1

Let  $(X, d)$  be a metric space. A subset  $E \subset X$  is called *dense* in  $E$  if, for any open  $O \subset X$ , we have  $O \cap E \neq \emptyset$ . A metric space  $(X, d)$  is called *separable* if it has a countable dense subset.

- (a) Show that  $\mathbb{Q}$  is dense in  $\mathbb{R}$ , and conclude that  $\mathbb{R}$  is separable.
- (b) Show that a subset  $E$  is dense in  $X$  if and only if  $\overline{E} = X$ .
- (c) By following these steps, prove that any compact metric space  $(X, d)$  is separable:
  - (i) Using problem 5(c) on homework 2, show that for each  $n \in \mathbb{N}$ , the following is true: there is a finite set  $S_n \subset X$  so that every point of  $X$  is within distance  $\frac{1}{n}$  of some point of  $S_n$ .
  - (ii) Let  $S = \bigcup_{n=1}^{\infty} S_n$ . Show that  $S$  is countable.
  - (iii) Show that every point of  $X$  is either in  $S$  or a limit point of  $S$ . Conclude that  $S$  is dense in  $X$ , and hence that  $X$  is separable.

## 1.1 Solution (a)

**Alexis's Proposition:** For any  $a, b \in \mathbb{R}$  where  $a < b$ , there exists some  $\frac{m}{n} \in \mathbb{Q}$  such that  $a < \frac{m}{n} < b$ .

*Proof.* Let  $a, b \in \mathbb{R}$  where  $a < b$ . Since  $a < b$ ,  $b - a > 0$ . By the Archimedean property, there exists some  $n \in \mathbb{N}$  such that  $n(b - a) > 1$ . Distribute and add  $na$  to both sides to find  $nb > na + 1$ . Also by the Archimedean property, there exists some natural numbers  $m_1$  and  $m_2$  such that  $m_1 > na$  and  $m_2 > -na$ . The latter means that  $na > -m_2$ . Therefore,  $m_1 > na > -m_2$ . Since  $m_1$  and  $-m_2$  are both integers, there exists some integer  $m$  such that  $-m_2 < m < m_1$  and  $m - 1 \leq na < m$ . This is equivalent to  $m \leq na + 1 < m + 1$ . Combine all of these and we find that  $na < m \leq na + 1 < nb$ . Shortening this,  $na < m < nb$ . Divide this all by  $n$  to find  $a < \frac{m}{n} < b$ . Since  $n$  is a natural number and  $m$  is an integer,  $\frac{m}{n}$  is a rational number. Ergo, if  $a, b \in \mathbb{R}$  and  $a < b$ , there exists some  $\frac{m}{n} \in \mathbb{Q}$  such that  $a < \frac{m}{n} < b$ . OEΔ

Now we prove  $\mathbb{Q}$  is dense in  $\mathbb{R}$  with the usual metric.

*Proof.* Take any set  $O$  open in  $\mathbb{R}$ . This means that for any  $o \in O$ , there exists some  $r > 0$  such that there exists an open ball  $B_r(o)$  that is a subset of  $O$ . We define this open ball as  $B_r(o) = \left\{ x \in \mathbb{R} \mid |x - o| < r \right\}$ .

Since  $r > 0$ , this means  $r > -r$  and therefore  $o + r > o - r$ . By Alexis's Proposition, there exists some rational number  $\frac{m}{n}$  such that  $o - r < \frac{m}{n} < o + r$ . Subtract  $o$  from all sides to find  $-r < \frac{m}{n} - o < r$ . By the definition of the absolute value,  $\left| \frac{m}{n} - o \right| < r$ . Therefore,  $\frac{m}{n} \in B_r(o)$  and by extension  $\frac{m}{n} \in O$ . Therefore,  $\frac{m}{n} \in \mathbb{Q} \cap O$ . Therefore, OEΔ

## 1.2 Solution (b)

First, we prove that if  $E$  is dense in  $X$ , then  $\overline{E} = X$ .

*Proof.* Suppose that  $E$  is dense in  $X$ . Take any point  $x \in X$ . Create an open ball of radius  $r$  for any  $r > 0$ . We can define the open ball as  $B_r(x) = \{y \in X \mid |y - x| < r\}$ . Since open balls are open and  $E$  is dense in  $X$ , there is a point  $p \in E \cap B_r(x)$ . Therefore, there is a point  $p$  in  $E$  in every open ball centered at any point in  $X$ . Therefore, every point in  $X$  is a limit point of  $E$ . Therefore,  $X$  is the set of limit points of  $E$ . Since  $\overline{E}$  is the union of  $E$  and the set of its limit points and every point in  $E$  is in  $X$  (since  $E$  is a subset of  $X$ ),  $X = \overline{E}$ . OEΔ

Next we prove that  $E$  is dense in  $X$  if  $\overline{E} = X$ .

*Proof.* Let  $\overline{E} = X$ . Take any open set  $O \subset X$ . Since  $\overline{E} = E \cup E'$ , for any point  $p \in X$ , either  $p \in E$  or  $p \in E'$ . OEΔ

### 1.3 Solution (c)

## 2 Problem 2

Using **only** the definitions of  $\lim$ ,  $\limsup$ , and  $\liminf$ , prove the following statements.

(a)  $\lim_{n \rightarrow \infty} [\sqrt{n+1} - \sqrt{n}] = 0.$     *Hint:* Use difference of squares.

(b)  $\limsup_{n \rightarrow \infty} (-2)^n = \infty.$

(c)  $\limsup_{n \rightarrow \infty} [-\sqrt{n}] = -\infty.$

(d)  $\liminf_{n \rightarrow \infty} \left[ \frac{1}{n} + (1 + (-1)^n)n \right] = 0.$

## 2.1 Solution (a)

## 2.2 Solution (b)

### 2.3 Solution (c)



## 2.4 Solution (d)

### 3 Problem 3

Let  $\{s_n\}$  be an arbitrary sequence in  $\mathbb{R}$ .

(a) Let  $s \in \mathbb{R}$ . Show that  $\limsup_{n \rightarrow \infty} s_n = s$  if and only if the following two conditions hold:

- For all  $\epsilon > 0$ , there is  $N \in \mathbb{N}$  so that  $s_n < s + \epsilon$  for all  $n \geq N$ .
- For all  $\epsilon > 0$  and for all  $N \in \mathbb{N}$ , there exists  $n \geq N$  so that  $s_n > s - \epsilon$ .

(b) Show that some subsequence of  $\{s_n\}$  converges (in the sense of extended real numbers, if necessary) to  $\limsup_{n \rightarrow \infty} s_n$  and some subsequence of  $\{s_n\}$  converges to  $\liminf_{n \rightarrow \infty} s_n$ .

(c) Define

$$E = \{x \in [-\infty, \infty] : \text{some subsequence } \{s_{n_k}\} \text{ of } \{s_n\} \text{ converges to } x\}.$$

Show that  $\limsup_{n \rightarrow \infty} s_n = \sup E$  and  $\liminf_{n \rightarrow \infty} s_n = \inf E$ .

*Hint:* what happens to  $\limsup$  and  $\liminf$  on subsequences?

### 3.1 Solution (a)

### 3.2 Solution (b)

### 3.3 Solution (c)

## 4 Problem 4

Let  $a > 1$ . In this problem, we will define what it means to write  $a^r$  for a positive *irrational* number  $r$ . We first define exponentiation for positive *integer* exponents in the usual way, via repeated multiplication. We know this satisfies some properties, for example

$$a^{r+s} = a^r a^s \quad (*)$$

$$(ab)^r = a^r b^r \quad (\Delta)$$

for  $a, b > 0$  and  $r, s > 0$  integers. Now, we proved in class that for any positive integer  $q$ , there is a *unique* positive number whose  $q$ th power is  $a$ , which we denote  $\sqrt[q]{a}$ . For a positive rational number  $r = \frac{p}{q}$ , we define  $a^r = \sqrt[q]{a^p}$ .

- Show that this definition makes sense, i.e. is independent of the representation of  $r$  as a fraction. That is, show that if rational numbers  $\frac{p}{q} = \frac{r}{s}$  then  $\sqrt[q]{a^p} = \sqrt[s]{a^r}$ . *Hint:* use lowest terms and the uniqueness of  $q$ th roots.
- Show that properties  $(*)$  and  $(\Delta)$  above still hold for positive rational numbers. *Hint:* find a common denominator and use uniqueness again.
- For positive rational  $r < s < t$ , show that  $a^r < a^s$  and that

$$0 < a^s - a^r = a^r (a^{s-r} - 1).$$

Conclude that for any positive rational  $r, s$  and any rational  $t$  larger than both  $r$  and  $s$ , we have

$$|a^r - a^s| < a^t (a^{|r-s|} - 1).$$

- Now let  $r$  be a positive *irrational* number. We know there is a sequence of rational numbers  $q_k$  that converge to  $r$  (for example, the decimal expansion of  $r$ ). We define

$$a^r = \lim_{k \rightarrow \infty} a^{q_k}.$$

Show that the sequence  $\{a^{q_k}\}$  is Cauchy, and conclude that this limit exists.

*Hint:* Use part (c) and Rudin Theorem 3.20(d).

- Show that the choice of sequence  $\{q_k\}$  is irrelevant; that is, show that if  $\{p_k\}$  and  $\{q_k\}$  are sequences of rational numbers converging to  $r$ , then

$$\lim_{k \rightarrow \infty} a^{p_k} = \lim_{k \rightarrow \infty} a^{q_k}.$$

- Using limit laws, prove that properties  $(*)$  and  $(\Delta)$  continue to hold for positive irrational exponents.

*Remark:* For  $r < 0$  we then define  $a^r = \frac{1}{a^{-r}}$ , and for  $0 < a < 1$  we define  $a^r = \left(\frac{1}{a}\right)^{-r}$ . It is then a matter of checking cases to verify that all the properties of exponentiation we want are indeed true.

#### 4.1 Solution (a)

**Billie's Proposition:** If  $\frac{p}{q} = \frac{r}{s}$  and  $\frac{p}{q}$  is in lowest terms but  $\frac{r}{s}$  is not, there exists some nonzero integer  $t$  such that  $r = pt$  and  $s = qt$ .

*Proof.* Let  $\frac{p}{q} = \frac{r}{s}$  and  $\frac{p}{q}$  is in lowest terms but  $\frac{r}{s}$  is not. Multiply by to find that  $ps = qr$ .

Divide by  $p$  to find that  $s = \frac{qr}{p}$ . Since  $s$  is a nonzero integer,  $p$  divides  $qr$ . Since  $\frac{p}{q}$  is in lowest terms,  $p$  does not divide  $q$ , so  $p$  must divide  $r$ . Therefore, there is some nonzero integer  $t$  such that  $p = rt$ .

Substitute in our value of  $p$  into the result of the cross multiplication to find  $rst = qr$ . Divide both sides by  $r$  to find that  $q = st$ . OEΔ

*Proof.* Let  $x = \sqrt[q]{a^p}$  and  $y = \sqrt[s]{a^r}$ , and suppose that  $\frac{p}{q} = \frac{r}{s}$  and that  $\frac{p}{q}$  is in lowest terms while  $\frac{r}{s}$  is not. This would mean that  $x = a^{\frac{p}{q}}$  and  $y = a^{\frac{r}{s}}$ . Take  $x$  to the power of  $q$  and  $y$  to the power of  $s$  to find  $x^q = a^p$  and  $y^s = a^r$ . Take the former to the power of  $r$  and the latter to the power of  $p$  to find  $x^{qr} = a^{pr}$  and  $y^{ps} = a^{pr}$ . Transistively, this means  $x^{qr} = y^{sp}$ .

By Billie's Proposition, since  $\frac{p}{q}$  is in lowest terms while  $\frac{r}{s}$  is not, there exists some nonzero integer  $t$  such that  $r = pt$  and  $s = qt$ . Substitute these into our equation to find that  $x^{qpt} = y^{qpt}$ .

$$x^{qr} = y^{sp} \tag{1}$$

$$x = y \tag{2}$$

$$(x^{qr})^{sp} = (y^{sp})^{qr} \tag{3}$$

$$x^{qr} = (y^{sp})^{\frac{q}{p} \cdot \frac{r}{s}} \tag{4}$$

OEΔ

## 4.2 Solution (b)



### 4.3 Solution (c)

#### 4.4 Solution (d)

#### 4.5 Solution (e)

#### 4.6 Solution (f)